

RFID BASED ATTENDANCE SYSTEM USING RTC MODULE

(MINI PROJECT REPORT)

Arduino-Based RFID Attendance Monitoring System with
Real-Time Clock (RTC)

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ABSTRACT

The RFID Based Attendance System Using RTC Module is an Arduino-based embedded system developed to automate the process of attendance recording. Traditional attendance methods are time-consuming, prone to human errors, and difficult to maintain. This project utilizes RFID (Radio Frequency Identification) technology to identify authorized users quickly and accurately through RFID cards or tags. Each user is assigned a unique RFID card, which is scanned using an RFID reader to mark attendance automatically.

The system is built around the Arduino Uno microcontroller, which receives the RFID card information and verifies it against predefined user IDs. A DS1307 Real-Time Clock (RTC) module is integrated to provide accurate date and time information, enabling attendance records to be timestamped. A 16×2 LCD display is used to show user information and attendance status, while LEDs and a buzzer provide visual and audible indications for valid and invalid card scans.

This project demonstrates the practical application of RFID technology, real-time clock modules, and embedded systems in attendance management. The system is simple, cost-effective, reliable, and suitable for educational institutions, offices, and organizations requiring automated attendance tracking. Future enhancements may include database integration, cloud-based storage, biometric authentication, and wireless communication for remote attendance monitoring.

INTRODUCTION

Attendance management is an essential task in educational institutions, offices, and organizations. Traditional attendance methods such as manual registers are time-consuming, prone to errors, and can be manipulated easily. To overcome these limitations, automated attendance systems using RFID technology have become increasingly popular.

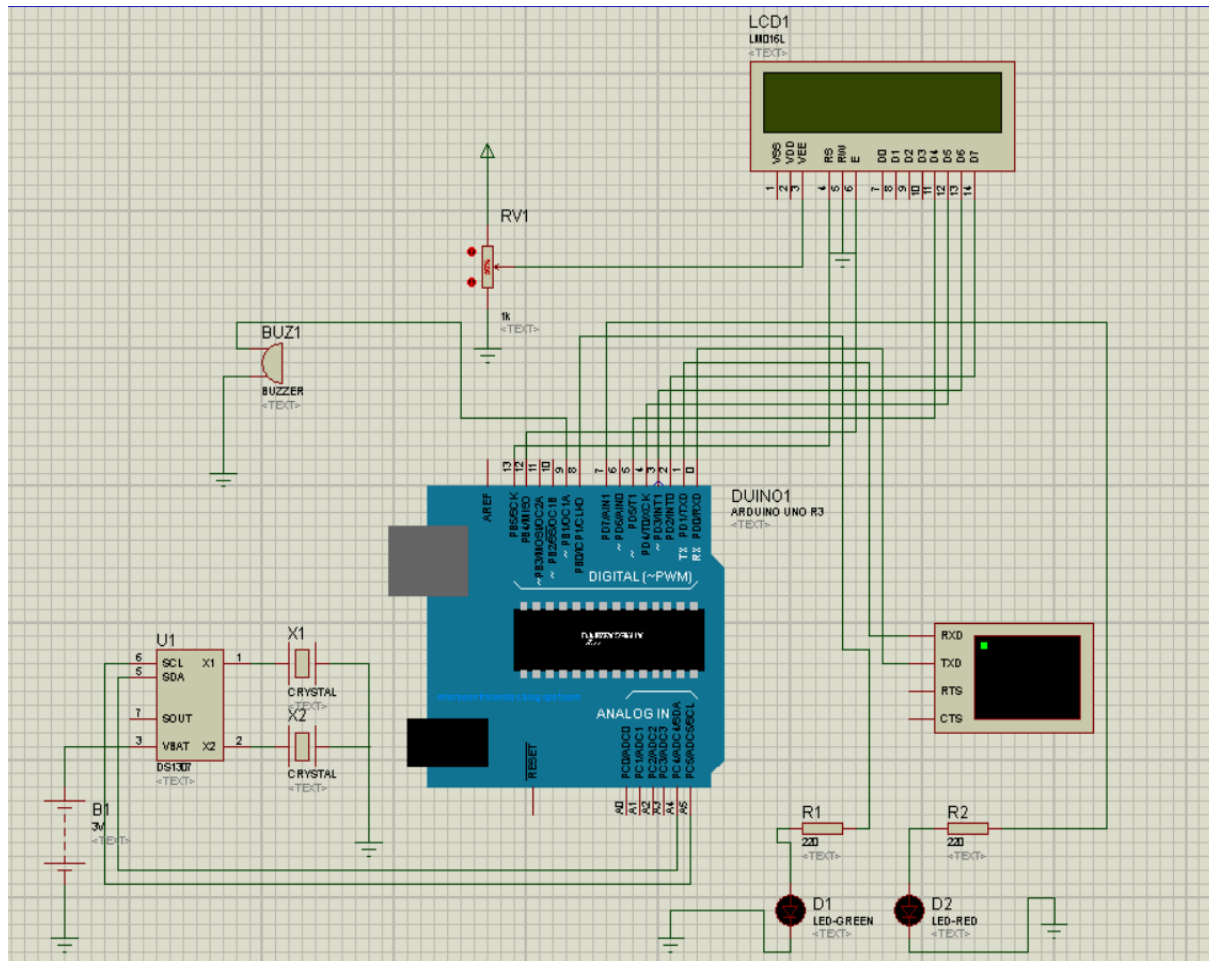
The RFID Based Attendance System Using RTC Module is an Arduino-based embedded system designed to automate attendance recording. The system uses RFID cards for user identification and a DS1307 Real-Time Clock (RTC) module to provide accurate date and time information. When a user scans a valid RFID card, the system verifies the card ID and records attendance along with the corresponding timestamp.

The project integrates Arduino Uno, RFID technology, RTC module, LCD display, LEDs, and a buzzer to create a reliable and user-friendly attendance management solution. This system improves accuracy, reduces manual effort, and provides real-time attendance monitoring for educational and organizational environments.

COMPONENTS USED

S:NO	COMPONENT	QUANTITY	PURPOSE
1.	Arduino UNO	1	Microcontroller used to process sensor data
2.	EM-18 RFID Readed	1	Reads RFID Card Information
3.	RFID Cards/Tags	2 or more	User Identification
4.	DS1307 RTC Module	1	User Identification
5.	16×2 LCD Display	1	Displays System Status
6.	Green LED	1	Valid Card Indication
7.	Red LED	1	Invalid Card Indication
8.	Buzzer	1	Audio Alert
9.	Potentiometer	1	LCD Contrast Adjustment
10.	Virtual Terminal	11	RFID Simulation in Proteus
11.	Crystal Oscillator	1	RTC Clock Source
12	Battery	1	RTC Backup Power

CIRCUIT DIAGRAM



DESCRIPTION

The RFID Based Attendance System is designed to automatically record attendance by identifying users through RFID cards. Each RFID card contains a unique identification number. When a card is scanned, the RFID reader sends the card ID to the Arduino Uno.

The Arduino compares the received ID with predefined authorized IDs stored in the program. If the card is valid, the attendance is marked and the DS1307 RTC module provides the current date and time for timestamping. The attendance information is displayed on the LCD screen and logged through the serial terminal.

For valid cards, the green LED glows and the buzzer provides a confirmation beep. If an invalid card is scanned, the red LED glows and an access denied message is displayed.

WORKING PRINCIPLE

The system operates based on RFID identification and real-time clock synchronization.

1. The RFID card is placed near the RFID reader.
2. The RFID reader detects and reads the card UID.
3. The UID is transmitted to the Arduino Uno through serial communication.
4. The Arduino compares the received UID with stored authorized IDs.
5. The DS1307 RTC module provides the current date and time.
6. If the UID is valid:
 - Attendance is recorded.
 - Date and time are attached to the attendance entry.
 - LCD displays attendance confirmation.
 - Green LED and buzzer are activated.
7. If the UID is invalid:
 - Attendance is rejected.
 - LCD displays an invalid card message.
 - Red LED is activated.
8. The system returns to waiting mode for the next card scan.

ARDUINO IDE CODE

```
#include <Wire.h>
#include <RTCLib.h>
#include <LiquidCrystal.h>

#define GREEN_LED 8
#define RED_LED 7
#define BUZZER 9

LiquidCrystal lcd(13, 12, 5, 4, 3, 2);
RTC_DS1307 rtc;

String cardID = "";

void setup()
{
  Serial.begin(9600);

  lcd.begin(16, 2);

  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(BUZZER, OUTPUT);

  if (!rtc.begin())
  {
    lcd.print("RTC Error");
    while (1);
  }

  // Run once to set RTC time, then comment it out
  // rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));

  lcd.print("RFID ATTENDANCE");
  lcd.setCursor(0, 1);
  lcd.print("System Ready");
  delay(2000);

  lcd.clear();
  lcd.print("Scan Card...");
}
```



```

void loop()
{
  if (Serial.available())
  {
    cardID = Serial.readStringUntil('\n');
    cardID.trim();
    DateTime now = rtc.now();
    lcd.clear();
    if (cardID == "1234")
    {
      lcd.print("Student 1");
      lcd.setCursor(0, 1);
      lcd.print("Present");

      digitalWrite(GREEN_LED, HIGH);

      tone(BUZZER, 1000);
      delay(300);
      noTone(BUZZER);

      Serial.println();
      Serial.println("===== ATTENDANCE =====");
      Serial.println("ID: 1234");
      Serial.println("Name: Student 1");

      Serial.print("Date: ");
      Serial.print(now.day());
      Serial.print("/");
      Serial.print(now.month());
      Serial.print("/");
      Serial.println(now.year());

      Serial.print("Time: ");
      Serial.print(now.hour());
      Serial.print(":");
      Serial.print(now.minute());
      Serial.print(":");
      Serial.println(now.second());

      Serial.println("Status: Present");
      Serial.println("=====");

      delay(2000);
    }
  }
}

```

```

    digitalWrite(GREEN_LED, LOW);
}
else if (cardID == "5678")
{
    lcd.print("Student 2");
    lcd.setCursor(0, 1);
    lcd.print("Present");

    digitalWrite(GREEN_LED, HIGH);

    tone(BUZZER, 1000);
    delay(300);
    noTone(BUZZER);

    Serial.println();
    Serial.println("===== ATTENDANCE =====");
    Serial.println("ID: 5678");
    Serial.println("Name: Student 2");

    Serial.print("Date: ");
    Serial.print(now.day());
    Serial.print("/");
    Serial.print(now.month());
    Serial.print("/");
    Serial.println(now.year());

    Serial.print("Time: ");
    Serial.print(now.hour());
    Serial.print(":");
    Serial.print(now.minute());
    Serial.print(":");
    Serial.println(now.second());

    Serial.println("Status: Present");
    Serial.println("=====");

    delay(2000);
    digitalWrite(GREEN_LED, LOW);
}
else
{
    lcd.print("Invalid Card");
    lcd.setCursor(0, 1);
    lcd.print("Access Denied");
}

```

```
digitalWrite(RED_LED, HIGH);

tone(BUZZER, 500);
delay(700);
noTone(BUZZER);

Serial.println();
Serial.println("==== ATTENDANCE =====");
Serial.print("Unknown ID: ");
Serial.println(cardID);
Serial.println("Status: Denied");
Serial.println("=====");

delay(2000);
digitalWrite(RED_LED, LOW);
}

lcd.clear();

lcd.print(now.hour());
lcd.print(":");

if (now.minute() < 10)
    lcd.print("0");

lcd.print(now.minute());

lcd.setCursor(0, 1);

lcd.print(now.day());
lcd.print("/");

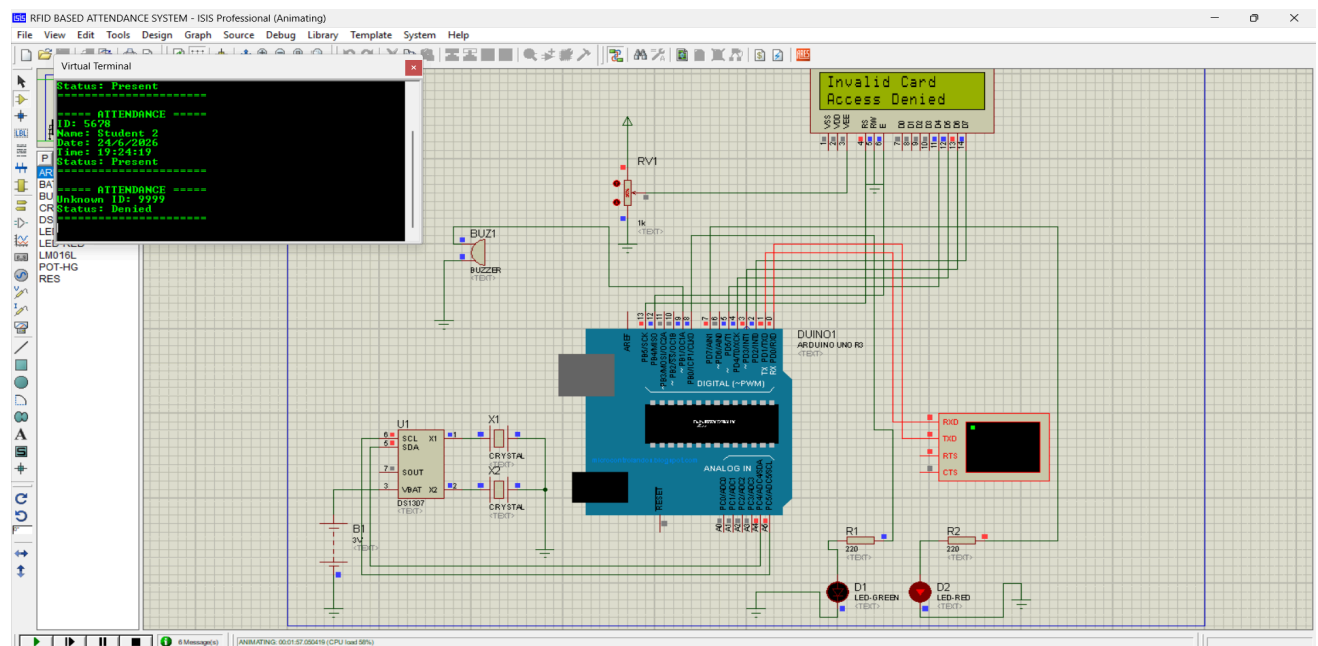
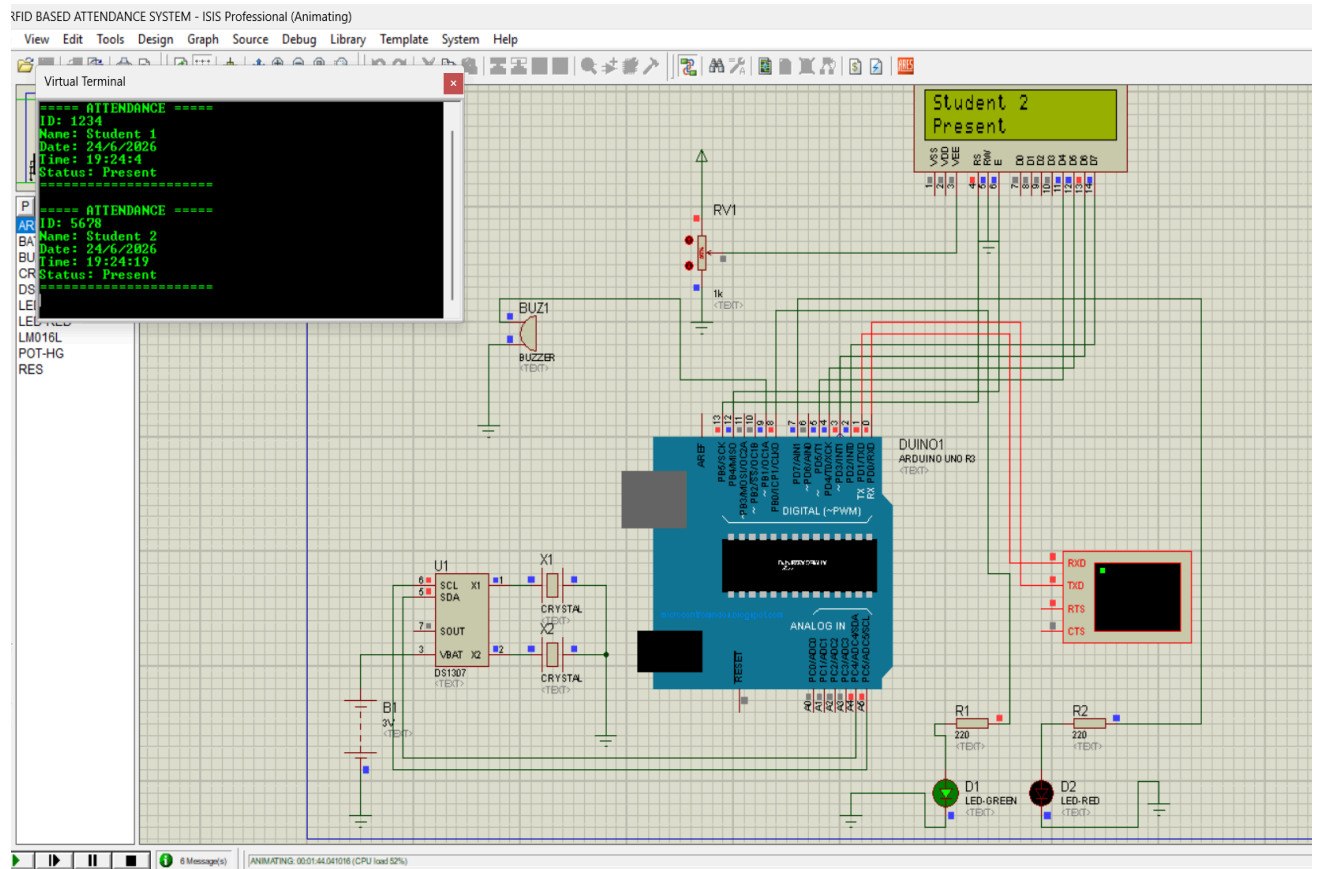
lcd.print(now.month());
lcd.print("/");

lcd.print(now.year());

delay(3000);

lcd.clear();
lcd.print("Scan Card...");
}
```

RESULT & OUTPUT



FUTURE SCOPE

The RFID Based Attendance System can be enhanced by integrating cloud databases for centralized attendance storage. Wireless communication technologies such as Wi-Fi, GSM, and Bluetooth can be added for remote monitoring. Biometric authentication can be combined with RFID for improved security.

Future versions may include mobile applications, web-based attendance dashboards, automatic report generation, facial recognition systems, and IoT-based attendance monitoring for smart institutions.

ADVANTAGES

- Fast and automatic attendance recording.
- Eliminates manual attendance errors.
- Accurate timestamping using RTC module.
- Low-cost implementation.
- Easy to operate and maintain.
- Improved security through RFID authentication.
- Real-time attendance monitoring.
- Suitable for schools, colleges, and offices.

CONCLUSION

The RFID Based Attendance System Using RTC Module was successfully developed using Arduino Uno, RFID technology, DS1307 RTC module, LCD display, LEDs, and a buzzer. The system automates attendance recording and accurately records the date and time of attendance events.

The project demonstrates the effective integration of RFID technology and embedded systems to create a reliable attendance management solution. It reduces manual effort, improves accuracy, and provides a cost-effective approach for attendance tracking in educational institutions and organizations. The system serves as a strong foundation for future smart attendance and access control applications.